CHAPTER 1.1 · INTRODUCTION TO ECONOMICS

Economics: the study of choice under **scarcity** — unlimited wants meet limited resources. Three core questions: **What** to produce? **How** to produce it? **For whom**?

Scarcity can also describe a market move — a fall in supply relative to demand pushing a new equilibrium price.

1.1.2 Opportunity Cost

Value of the next-best alternative foregone when a choice is made.

- » **Explicit costs:** actual cash outflows — measurable, recorded in accounts.
- » **Implicit costs:** no cash changes hands but real value is foregone (e.g. owner's unpaid time).
- » **Risk vs Opportunity Cost:** risk is loss of capital; opportunity cost is the return given up by choosing one investment over another.

> **CASE STUDY · The Most Expensive Pizza**
>
> In 2010, 10,000 BTC bought two pizzas (worth ~$41 then). By Aug 2024 the holding was worth over **US$690 million** — the textbook illustration of opportunity cost.

1.1.3 Market Price

The price at which **quantity supplied = quantity demanded**. Determined by the interaction of supply & demand.

- » **Consumer surplus:** max price a consumer would pay minus the market price actually paid.
- » **Producer surplus:** market price minus the producer's minimum acceptable price (cost).

1.1.5 / 1.1.6 Macro vs Micro

Macroeconomics	Microeconomics
Whole economy — GDP, inflation, unemployment, exchange rates, monetary & fiscal policy.	Individual firms, households & markets — supply, demand, prices, consumer & producer behaviour.

1.1.8–1.1.9 Theory: Assumptions & Schools

Economists build models on **simplifying assumptions** — rational agents, perfect info, ceteris paribus, profit / utility maximisation.

Neo-classical view: rational behaviour, free markets and competition drive efficient outcomes.

Positive Economics	Normative Economics
Objective, fact-based — describes **what IS**; testable against evidence.	Subjective, value-based — prescribes **what SHOULD BE**; opinion-driven.

1.2 THEORY OF DEMAND & SUPPLY

1.2.2 Supply Curve

Slopes **upward**: higher price → suppliers willing to supply more. The **independent variable (price) is plotted on the Y-axis** — exception to maths convention.

1.2.3 Demand Curve

Slopes **downward** (inverse): higher price → less demanded. Same Y-axis price convention.

1.2.4 Movement vs Shift

- » **Change in quantity demanded / supplied** = **movement ALONG** the curve, caused by price ONLY.
- » **Change in demand / supply** = **SHIFT** of the whole curve, caused by income, tastes, expectations, prices of related goods, input costs, technology, taxes / subsidies.

1.2.5 Market Equilibrium

Where supply & demand curves intersect — **Qs = Qd** at the equilibrium price.

- » **Price > equilibrium** → **surplus** (excess supply).
- » **Price < equilibrium** → **shortage** (excess demand).

Price Elasticity (PED / PES)

% change in quantity for a 1% change in price.

Value	Interpretation
> 1	Elastic — quantity highly responsive.
< 1	Inelastic — quantity barely changes (necessities, fuel).
= 1	Unit elastic.
= 0	Perfectly inelastic — vertical curve.
= ∞	Perfectly elastic — horizontal curve.

1.2.7 Price Ceiling vs Price Floor

Tool	Effect
Price ceiling max — set BELOW equilibrium	Shortage; rationing, black markets, lower investment. e.g. rent control.
Price floor min — set ABOVE equilibrium	Surplus; unsold inventory or unemployment. e.g. minimum wage, ag price support.

MEMORISE · The 3 Rules of Curves

1. Price on Y-axis (exception to maths convention).
2. Movement along curve = price-only change.
3. Shift of curve = any other factor changes.

1.3 MARKET STRUCTURES

Defined by **number of firms, product differentiation, barriers to entry, pricing power** and access to information.

Structure	Key Features
Perfect Competition	Many small firms, identical products, price takers, free entry / exit, perfect info. Zero economic profit long-run.
Monopolistic Competition	Many firms, differentiated products, some pricing power, low entry barriers (e.g. restaurants, retail).
Oligopoly	Few large firms, identical or differentiated products, high barriers, strategic interdependence, may collude.
Monopoly	Single firm, no close substitutes, price maker, very high barriers, regulated where it persists.

1.4 BEHAVIOURAL ECONOMICS

Combines economics with psychology — real people deviate from the rational-agent ideal due to **heuristics, biases, emotions** and social context.

Traditional vs Behavioural Finance

Traditional	Behavioural
Rational utility maximisers, perfect info, mean-variance optimisation (Markowitz, CAPM), efficient markets.	Bounded rationality; biases (overconfidence, anchoring, herding, loss aversion, mental accounting, framing). Explains bubbles & momentum.

1.4.6 Utility Theory

Consumers maximise **utility** (satisfaction) given prices and budget; subject to **diminishing marginal utility** — each extra unit adds less satisfaction.

1.4.7 Efficient Market Hypothesis (EMH)

Form	Information Reflected
Weak	All historical price & volume data → technical analysis cannot consistently beat the market.
Semi-strong	All public information → fundamental analysis cannot consistently beat the market.
Strong	All info incl. private / insider → no one can consistently beat the market.

Criticisms: persistent anomalies, asset bubbles and behavioural patterns all challenge especially the strong form.

1.5 MACRO ECONOMICS

Whole-economy lens: inflation, unemployment, growth, exchange rates, balance of payments, fiscal & monetary policy.

1.5.1 Schools of Macro Thought

- **Classical:** markets self-correct, prices & wages flexible.
- **Keynesian:** demand drives output; active fiscal policy in recessions.
- **Monetarist:** money supply growth is the key driver of inflation (Friedman).
- **Neo-classical & Neo-Keynesian:** blend rational expectations with short-run rigidities.
- **Supply-side:** tax cuts & deregulation to expand productive capacity.

Limitations of macro: many variables, simplifying assumptions, long & variable time-lags, political interference.

1.5.2 Economic Growth — Preconditions

- Human capital (skills, education)
- Physical capital & infrastructure
- Technology & productivity (TFP)
- Natural resources
- Stable institutions, rule of law, policy certainty

1.5.3 Components of GDP (Expenditure)

> FORMULA · MEMORISE
>
> **GDP (Y) = C + I + G + (X – M)**
>
> C = household Consumption · I = business Investment · G = Government spending · X–M = net exports.

Income approach: GDP = wages + rents + interest + profits.

Y = C + S — Income equals Consumption + Savings.

MPC (Marginal Propensity to Consume) = fraction of each extra rand of income spent.

1.5.4 Nominal vs Real GDP

Measure	Definition
Nominal GDP	Output at **current** prices — includes inflation.
Real GDP	Output at **constant base-year** prices — true measure of volume.
GDP Deflator	Nominal / Real × 100 → economy-wide price level.

1.5.5 Growth Rate Formula

> FORMULA
>
> **Growth = $(Y_1 - Y_0) / Y_0 \times 100$**
>
> Real growth uses real GDP — strips out inflation.

1.6 THE BUSINESS CYCLE

Repetitive fluctuations in real GDP around its long-run potential path.

- » **Expansion** — rising output, falling unemployment.
- » **Peak** — economy operating above potential; inflationary pressure builds.
- » **Contraction / recession** — falling output (two consecutive quarters of negative GDP growth).
- » **Trough** — low point; idle capacity, high unemployment.
- » **Recovery** — output picks up back toward potential.

Output gap = Actual – Potential GDP. Positive → inflationary; negative → deflationary & unemployment.

1.7 UNEMPLOYMENT

Type	Cause
Frictional	Between jobs — voluntary, short-term.
Structural	Skills / location mismatch with available jobs.
Cyclical	Recession-driven downturn in demand for labour.
Seasonal	Predictable seasonal patterns (tourism, agriculture).

Natural rate = frictional + structural. Policy generally targets cyclical unemployment.

1.8 INFLATION

Sustained rise in the general price level — erodes purchasing power, distorts saving & investment, drives wage demands.

1.8.1 Consumer Price Index (CPI)

FORMULA · CPI

CPI = (Cost of basket at current prices / Cost of basket at base-year prices) × 100

Other measures: PPI (producer prices), GDP deflator (economy-wide).

1.8.2 Types of Inflation & Effects

Type	Driver / Impact
Demand-pull	Too much demand vs supply — firms can pass costs, equities may benefit short term.
Cost-push	Input cost rises (oil, wages, FX) — squeezes margins, bad for equities.
Built-in	Wage-price spiral — entrenched expectations.

1.8.3 MONETARY POLICY

Central bank (SARB) manages money supply & interest rates to anchor inflation and support balanced growth.

Tools

- » **Open Market Operations (OMO)** — buy / sell government bonds to add / drain liquidity.
- » **Reserve requirements** — minimum % of deposits banks must hold.
- » **Repo / discount rate** — rate at which SARB lends to commercial banks; the headline policy signal.
- » **Quantitative easing** — large-scale asset purchases when rates approach zero.
- » **Forward guidance** — communicating the future path of rates to anchor expectations.

Stance

Stance	Effect
Expansionary	Lower rates & more liquidity → cheaper credit → spend, invest, grow → inflation.
Contractionary	Higher rates & less liquidity → cool demand → reduce inflation, slow growth.

Limits: long & variable lags, liquidity trap at the zero lower bound, blunt against supply-side shocks.

1.8.4 FISCAL POLICY

National Treasury uses **government spending (G)** and **taxation (T)** to influence aggregate demand.

Stance	Effect
Expansionary	↑ G or ↓ T → stimulates demand, growth, employment. Funded by borrowing → deficit.
Contractionary	↓ G or ↑ T → cools demand, slows growth, reduces deficit.

» **Budget deficit** (G > T) funded by issuing bonds → can lift yields, risk crowding out private investment.

» **Budget surplus** (G < T) reduces public debt, may slow short-run demand.

» **Discretionary** (deliberate policy changes) vs **automatic stabilisers** (welfare, progressive tax).

» **Limits:** political delays, debt sustainability, crowding-out, time-lags in implementation.

EXAM FOCUS · Monetary vs Fiscal

Monetary: fast, blunt, controlled by independent central bank — works via rates & credit.

Fiscal: slower, more targeted, controlled by Treasury / political process — works via spending & tax.

KEY ACRONYMS & FORMULAS

Term	Meaning
GDP	Gross Domestic Product = C + I + G + (X – M)
GNP	Gross National Product — incl. net income from abroad
CPI	Consumer Price Index — basket-based inflation measure
PPI	Producer Price Index — input-side inflation
MPC	Marginal Propensity to Consume / Monetary Policy Committee
TFP	Total Factor Productivity
PED / PES	Price Elasticity of Demand / Supply
EMH	Efficient Market Hypothesis (Weak / Semi / Strong)
OMO	Open Market Operations (monetary tool)
SARB	South African Reserve Bank — central bank
BoP	Balance of Payments — current + capital & financial

KEY FORMULAS · ONE-PAGE RECAP

GDP (expenditure) = **C + I + G + (X – M)**

Y = C + S

Real GDP = Nominal GDP / GDP Deflator × 100

Growth rate = $(Y_1 - Y_0) / Y_0 \times 100$

CPI = (Cost current basket / Cost base-year basket) × 100

Inflation rate = $(CPI_1 - CPI_0) / CPI_0 \times 100$

PED = $\% \Delta Q_d / \% \Delta P$